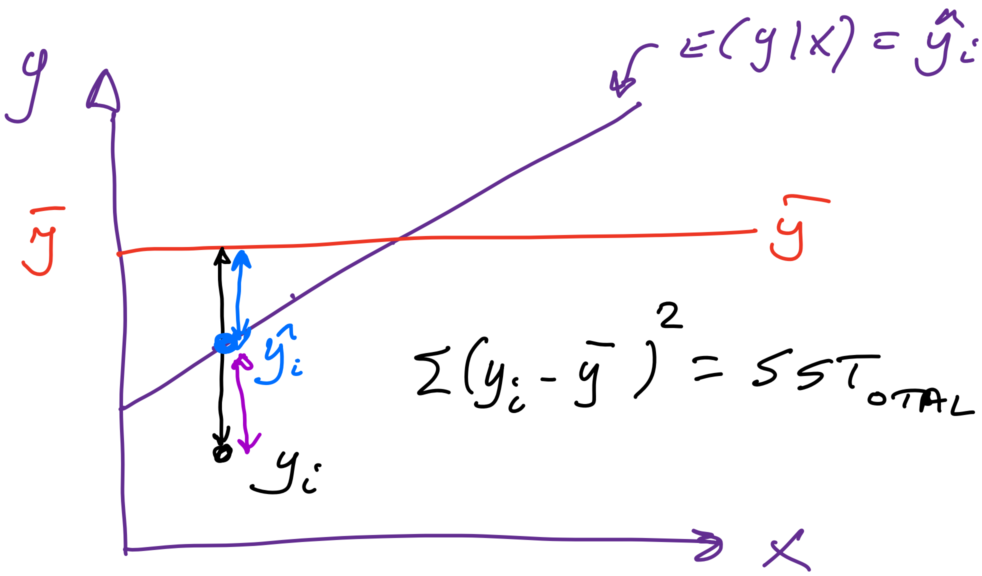


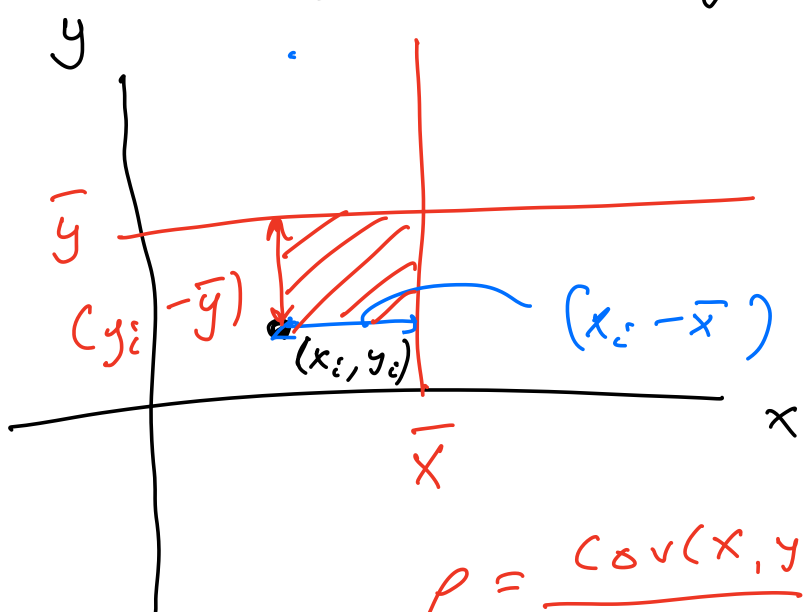
$$\Sigma(y_i - \bar{y})^2 = SST_{OTAL}$$

$SST = SSR + SSE$

$$\Sigma(y_i - \hat{y}_i)^2 = SSR_{egression}$$

$$\Sigma(\hat{y}_i - y_i)^2 = SSE_{RROR}$$

$$cov(x,y) = \Sigma(y_i - \bar{y})(x_i - \bar{x})$$



$$\rho = \frac{cov(x,y)}{\sigma_x \cdot \sigma_y}$$

DATA

$y = \beta_1 + \beta_2 \cdot x$

PARAMETERS $\hat{\beta}_2 = \dfrac{cov(x, y)}{var(x)}$